

Experiment: Crypt Arithmetic Problem

Aim

To solve a Crypt Arithmetic problem using Python by assigning unique digits to letters so that the arithmetic equation is satisfied.

Objectives

Understand the Crypt Arithmetic problem.

Apply brute-force search to find valid digit assignments.

Ensure each letter has a unique digit.

Display the correct solution.

Algorithm

1. Import the itertools module.
2. Define the letters used in the puzzle.
3. Generate all possible digit combinations.
4. Assign digits to each letter.
5. Check that leading letters are not zero.
6. Verify whether the arithmetic equation is true.
7. If a valid solution is found, display it.

8. Stop the program.

Pseudocode

```
START
Import itertools
Define letters
FOR each possible digit assignment
    IF leading digit is zero
        Continue
    Convert letters into numbers
    IF arithmetic equation is true
        Print solution
    EXIT
END FOR
STOP
```

Working Python Code

```
from itertools import permutations

letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')

for p in permutations(range(10), 8):
    S, E, N, D, M, O, R, Y = p

    if S == 0 or M == 0:
        continue

    send = 1000*S + 100*E + 10*N + D
    more = 1000*M + 100*O + 10*R + E
    money = 10000*M + 1000*O + 100*N + 10*E + Y

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
```

```
print("MONEY =", money)
break
```

Result

The Crypt Arithmetic puzzle is solved successfully by finding the correct digit assignment.

Sample Output

```
SEND = 9567
MORE = 1085
MONEY = 10652
```

Flowchart



